25901235 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECECT200 - CIRCUIT SIMULATION AND PCB DESIGN

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Copper thickness is measured in [U] / [CO1]

   a). mm b). cm c). Ounces d). Volts

2. Etching is a [U] / [CO2]

   a). Mechanical process b). Chemical process c). Electrical process d). Thermal process

3. Silkscreen contains [U] / [CO3]

   a). Component names b). Values c). Polarity marks d). All of these

4. Main concern in digital PCB design is [U] / [CO2]

   a). Power loss b). Signal integrity c). Color d). Board shape

5. Proteus is widely used for [U] / [CO3]

   a).Microcontroller simulation b). Web design c). Audio editing d). Database

6. Ground plane is used for [U] / [CO1]

a). Heating b). Noise reduction c). Decoration d). Power storage

7. In the PCB fabrication process, which of the following steps is carried out immediately after surface cleaning and before exposure? [U] / [CO2]

a). Drilling b). Photoresist application c). Etching d). Testing

8. Good layout improves [U] / [CO3]

a). Reliability b). Noise c). Cost only d). Size only

9. HF routing prefers [U] / [CO2]

a). Sharp corners b). Right angles c). Smooth curves d). Long traces

10. Manual routing gives [U] / [CO3]

a). Less control b). More control c). No control d). Automatic results

11. Single sided PCB is used for [U] / [CO1]

a). Complex circuits b). High speed circuits c). Simple circuits d). RF circuits

12. Etching process removes [U] / [CO2]

a). Copper b). Solder mask c). Silk d). Drill bits

13. Footprints are selected during [U] / [CO3]

a). Fabrication b). Design stage c). Testing d). Packaging

14. Flexible PCB material is [U] / [CO1]

a). FR-4 b). Ceramic c). Polyimide d). Paper

15. IPC-2221 relates to [U] / [CO2]

a). Soldering b). PCB design c). Assembly d). Testing

16. Major advantage of multilayer PCB is [U] / [CO1]

a). Large size b). Compact design c). Poor reliability d). Low speed

17. Thermal management deals with [U] / [CO2]

a). Voltage b). Heat dissipation c). Signal speed d). Noise

18. Incorrect footprint assignment in PCB design can lead to: [U] / [CO3]

a). Perfect PCB b). Assembly error c). Low noise d). Better routing

19. Important factor in PCB design [U] / [CO1]

a). Color b). Signal integrity c). Shape d). Label size

20. KiCad supports [U] / [CO3]

a). Only PCB b). Only schematic c). Schematic & PCB d). Only simulation

21. Example of SMD package [U] / [CO1]

a). DIP b). TO-220 c). SOIC d). Wire wound

22. Crosstalk can be reduced by [U] / [CO2]

a). Increasing trace length b). Proper spacing c). Thin traces d). Sharp bends

23. THT provides [U] / [CO3]

a). Strong mechanical joint b). Weak joint c). No strength d). Temporary joint

24. Automation reduces [U] / [CO2]

a). Precision b). Human error c). Cost only d). Speed

25. Pads in footprint represent [U] / [CO3]

a). Tracks b). Component leads c). Text d). Vias

26. Error in schematic causes [U] / [CO1]

a). Correct PCB b). Wrong connections c). Low cost d). Fast fabrication

27. Transmission lines are important in [U] / [CO2]

a). HF signals b). Low frequency c). DC circuits d). LED circuits

28. Gerber files are used for [U] / [CO3]

a). Simulation b). Manufacturing c). Testing d). Programming

29. Manual drilling is [U] / [CO2]

a). Highly accurate b). Slow c). Automated d). Laser based

30. Design report includes [U] / [CO3]

a). BOM b). DRC c). Errors d). All of these

31. Which is an active component? [U] / [CO1]

a). Resistor b). Capacitor c). Diode d). Inductor

32. Photoresist is sensitive to [U] / [CO2]

a). Heat b). Light c). Pressure d). Sound

33. Netlist errors cause [U] / [CO3]

a). Correct PCB b). Connection mismatch c). Low cost d). Better EMI

34. Analog circuits are sensitive to [U] / [CO2]

a). Noise b). Color c). Size d). Font

35. PCB assembly follows [U] / [CO3]

a). Design b). Fabrication c). Testing d). Packaging

[1x35 = 35]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

36. Discuss the advantages and applications of double sided PCBs. [U] / [CO1]

37. Explain the through-hole plating process and its role in electrical interconnection between PCB layers. [U] / [CO2]

38. Explain the workflow involved in creating a schematic and PCB layout using EasyEDA. [U] / [CO3]

39. Discuss EMI and its effects on PCB performance. [U] / [CO2]

40. Explain the SMT assembly process with suitable examples. [U] / [CO3]

[5x3 = 15]